

# Differential Equation

## 7. Second Order Differential Equation

A second-order ODE is called **linear** if it can be written

$$y'' + p(x)y' + q(x)y = r(x)$$

and **nonlinear** if it cannot be written in this form.

Homogeneous Linear ODEs of Second Order (RHS=0)

$$y'' + p(x)y' + q(x)y = 0$$

DE with constant Coefficients

$$y'' + p y' + q y = 0, \quad p \text{ and } q \text{ are constants.}$$

Method of solution: Write Auxiliary equation by putting  $\frac{d}{dx} = D$

$$D^2 y + p D y + q y = 0, (D^2 + p D + q)y = 0, D^2 + p D + q = 0$$

We will get two roots from the above quadratic equation.

Roots are real and distinct  $m_1$  and  $m_2$ :  $y = c_1 e^{m_1 x} + c_2 e^{m_2 x}$

**Example:**  $y'' - y = 0$

**Solution:**  $D^2 - 1 = 0, D = 1, 1$   $y = e^x$  and  $y = e^{-x}$  are solutions of the homogeneous linear differential equation.

Now try to verify that

(a)  $y = e^x$

(b)  $y = e^{-x}$

(c)  $y = C_1 e^x + C_2 e^{-x}$

All of them are solutions of given ODE and hence they satisfy them.

Roots are real but same  $m_1, m_2 = m$ ;  $y = (c_1 + c_2 x) e^{mx}$

**Example:**  $y'' + 2y + 1 = 0$

**Solution:**  $(D+1)^2 = 0, D = -1, -1$   $y = e^{-x}$  and  $y = x e^{-x}$  are solutions of the homogeneous linear differential equation.

Now try to verify that

(a)  $y = e^{-x}$

(b)  $y = x e^{-x}$

(c)  $y = (C_1 + C_2 x) e^{-x}$

(d) Roots are complex  $\alpha \pm i\beta$ ;  $y = e^{mx} (A \cos \beta x + B \sin \beta x)$

**Example:**  $y'' + y = 0$ .

**Solution:**  $D^2 + 1 = 0, D = i, -i$   
 $y = e^{0x} (A \cos x + B \sin x)$

We may verify that  $y_1 = \cos x$  and  $y_2 = \sin x$  are solutions of the differential equation  $y'' + y = 0$ .

**Note:** Now  $\cos x$  and  $\sin x$  are not proportional,  $y_1 / y_2 = \cot x$  const.

Hence, they form a basis of our equation.

(i) Cauchy-Euler Form

$$x^2 y'' + x p y' + q y = 0 \quad p \text{ and } q \text{ are constants.}$$

Method of solution: Put  $x = e^z$ ,  $x \frac{d}{dx} = \frac{d}{dz} = D$

$D(D-1) + PD + Q = 0$  Calculate the roots and write the solution in same way as in the above case.

**Example:** Solve the Euler-Cauchy equation

$$x^2 y'' - 2.5 x y' - 2.0 y = 0.$$

**Solution:** The auxiliary equation is  $x \frac{d}{dx} = \frac{d}{dz} = D$ ,  $D(D-1) - 2.5PD - 2 = 0$

$D^2 - 3.5D - 2.0 = 0$ , The roots are  $D_1 = -0.5$  and  $D_2 = 4$ . This gives  $y_1 = e^{-0.5z}$ ,  $y_2 = e^{4z}$ ,

Substituting  $x = e^z$ ,  $y_1 = 1/\sqrt{x}$ ,  $y_2 = x^4$

and the general solution  $y = \frac{c_1}{\sqrt{x}} + c_2 x^4$  valid for all positive  $x$ .